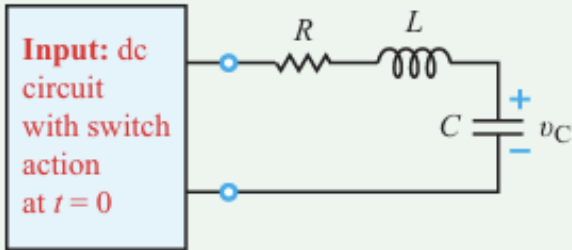
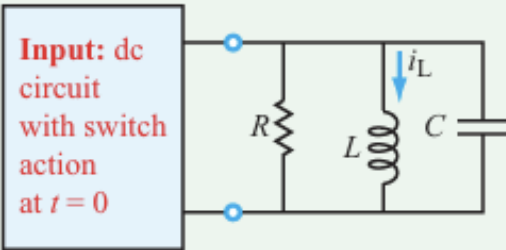


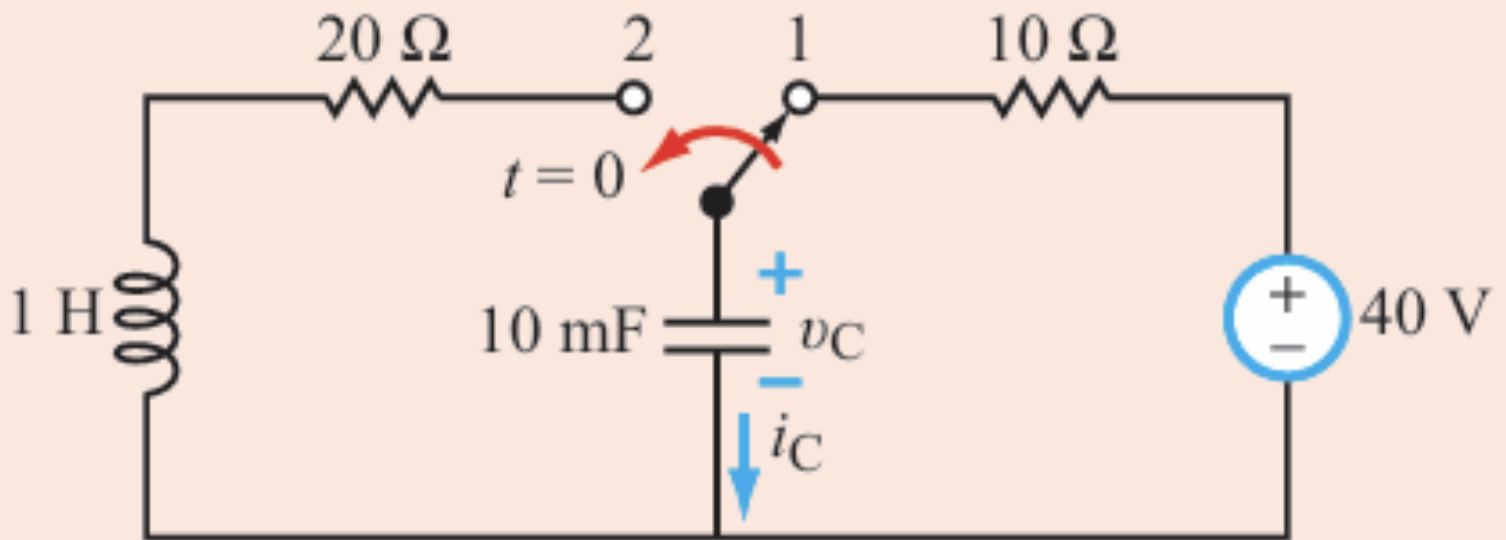
**Table 6-2: General solution for second-order circuits for  $t \geq 0$ .**

|                                                                                       |                                                                         |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| $x(t)$ = unknown variable (voltage or current)                                        |                                                                         |
| <b>Differential equation:</b>                                                         | $x'' + ax' + bx = c$                                                    |
| <b>Initial conditions:</b>                                                            | $x(0)$ and $x'(0)$                                                      |
| <b>Final condition:</b>                                                               | $x(\infty) = \frac{c}{b}$                                               |
| $\alpha = \frac{a}{2}$                                                                | $\omega_0 = \sqrt{b}$                                                   |
| <b>Overdamped Response</b> $\alpha > \omega_0$                                        |                                                                         |
| $x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$                             |                                                                         |
| $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$                                        | $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$                          |
| $A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$                               | $A_2 = -\left[ \frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$ |
| <b>Critically Damped</b> $\alpha = \omega_0$                                          |                                                                         |
| $x(t) = [(B_1 + B_2 t) e^{-\alpha t} + x(\infty)] u(t)$                               |                                                                         |
| $B_1 = x(0) - x(\infty)$                                                              | $B_2 = x'(0) + \alpha[x(0) - x(\infty)]$                                |
| <b>Underdamped</b> $\alpha < \omega_0$                                                |                                                                         |
| $x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t) e^{-\alpha t} + x(\infty)] u(t)$ |                                                                         |
| $D_1 = x(0) - x(\infty)$                                                              | $D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$               |
| $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$                                             |                                                                         |

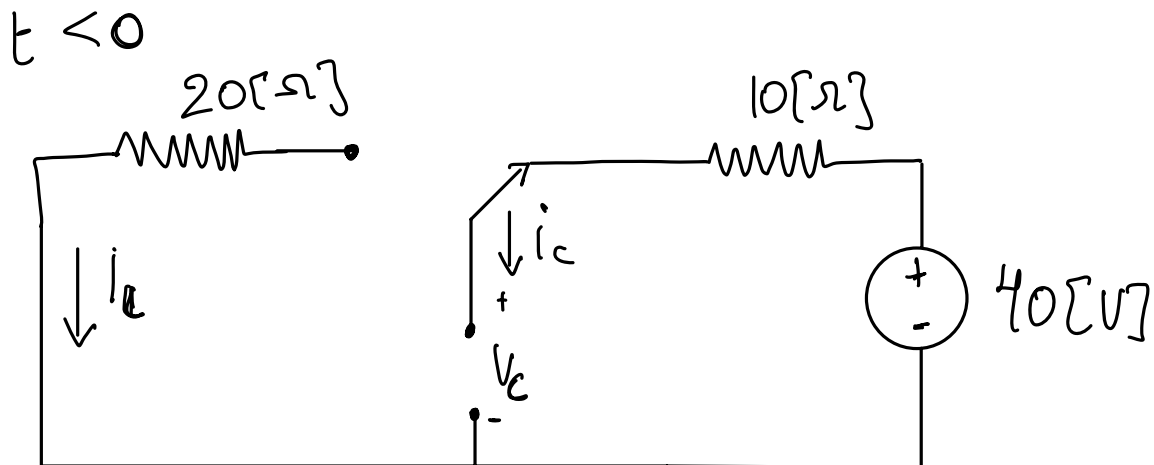
**Table 6-1:** Step response of RLC circuits for  $t \geq 0$ .

| <p style="text-align: center;"><b>Series RLC</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p style="text-align: center;"><b>Parallel RLC</b></p>                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p style="text-align: center;">Total Response</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p style="text-align: center;">Total Response</p>                                                                                                                                                                                                                       |
| <p><b>Overdamped</b> (<math>\alpha &gt; \omega_0</math>)</p> $v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$ $A_1 = \frac{\frac{1}{C} i_C(0) - s_2 [v_C(0) - v_C(\infty)]}{s_1 - s_2}$ $A_2 = \frac{\frac{1}{C} i_C(0) - s_1 [v_C(0) - v_C(\infty)]}{s_2 - s_1}$                                                                                                                                                                                                                                                                                                                                                                  | <p><b>Overdamped</b> (<math>\alpha &gt; \omega_0</math>)</p> $i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$ $A_1 = \frac{\frac{1}{L} v_L(0) - s_2 [i_L(0) - i_L(\infty)]}{s_1 - s_2}$ $A_2 = \frac{\frac{1}{L} v_L(0) - s_1 [i_L(0) - i_L(\infty)]}{s_2 - s_1}$ |
| <p><b>Critically Damped</b> (<math>\alpha = \omega_0</math>)</p> $v_C(t) = (B_1 + B_2 t) e^{-\alpha t} + v_C(\infty)$ $B_1 = v_C(0) - v_C(\infty)$ $B_2 = \frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]$                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>Critically Damped</b> (<math>\alpha = \omega_0</math>)</p> $i_L(t) = (B_1 + B_2 t) e^{-\alpha t} + i_L(\infty)$ $B_1 = i_L(0) - i_L(\infty)$ $B_2 = \frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]$                                                           |
| <p><b>Underdamped</b> (<math>\alpha &lt; \omega_0</math>)</p> $v_C(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$ $D_1 = v_C(0) - v_C(\infty)$ $D_2 = \frac{\frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]}{\omega_d}$                                                                                                                                                                                                                                                                                                                                                                                | <p><b>Underdamped</b> (<math>\alpha &lt; \omega_0</math>)</p> $i_L(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$ $D_1 = i_L(0) - i_L(\infty)$ $D_2 = \frac{\frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]}{\omega_d}$               |
| <p style="text-align: center;"><b>Auxiliary Relations</b></p> <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math display="block">\alpha = \begin{cases} \frac{R}{2L} &amp; \text{Series RLC} \\ \frac{1}{2RC} &amp; \text{Parallel RLC} \end{cases}</math> <math display="block">s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}</math> </div> <div style="width: 45%;"> <math display="block">\omega_0 = \frac{1}{\sqrt{LC}}</math> <math display="block">\omega_d = \sqrt{\omega_0^2 - \alpha^2}</math> <math display="block">s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}</math> </div> </div> |                                                                                                                                                                                                                                                                         |

**Exercise 6-4:** The switch in **Fig. E6.4** is moved to position 2 after it had been in position 1 for a long time. Determine: (a)  $v_C(0)$  and  $i_C(0)$ , and (b)  $i_C(t)$  for  $t \geq 0$ .



**Figure E6.4**

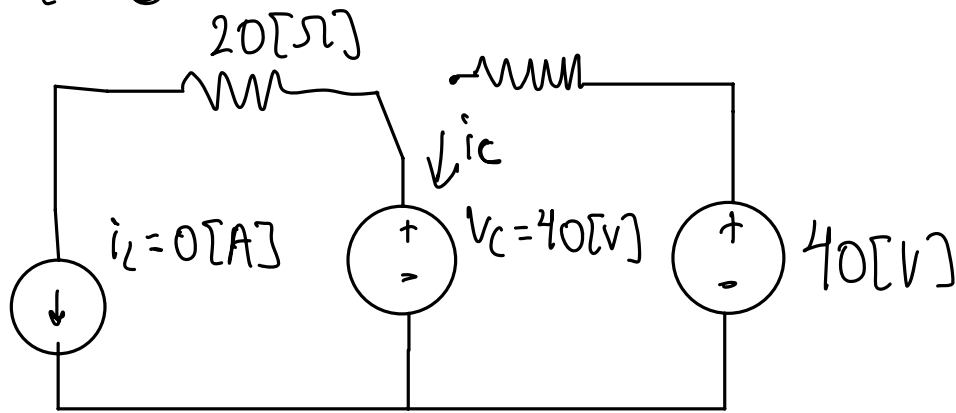


$$i_L(0^-) = i_L(0) = 0 \text{ [A]}$$

$$i_C(0^-) = 0 \text{ [A]}$$

$$V_C(0^-) = V_C(0^+) = 40 \text{ [V]}$$

at  $t = 0$



$$i_L(0) = i_C(0) = 0[A]$$

we have series RLC

$$\text{so } \alpha = \frac{R}{2L} = \frac{20}{2 \times 1} = 10 [Np/s]$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10 \times 10^{-3}}} = 10 [\text{rad/s}]$$

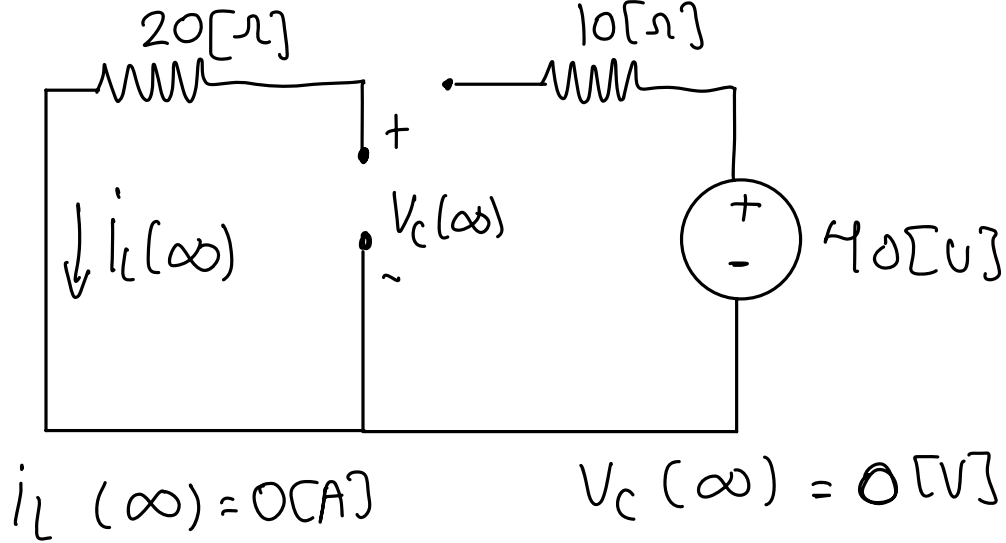
so it is

Critically damped

$$V_C(t) = (B_1 + B_2 t) e^{-\alpha t} + V_C(\infty)$$

$$B_1 = V_C(0) - V_C(\infty)$$

$$B_2 = \frac{1}{C} i_C(0) + \alpha [V_C(0) - V_C(\infty)]$$



$$B_1 = 40$$

$$B_2 = 400$$

$$V_C(t) = (40 + 400t)e^{-10t} \quad t \geq 0$$

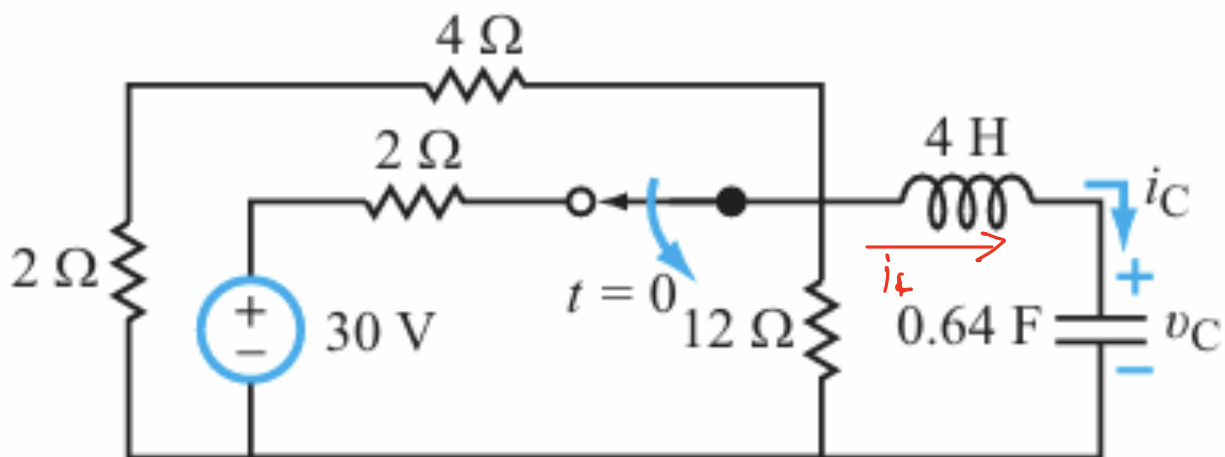
$$i_C(t) = C \frac{dV_C(t)}{dt}$$

$$= 10 \text{ [mF]} \left( 400e^{-10t} + (-10e^{-10t})(40 + 400t) \right)$$

$$= 10 \text{ [mF]} \left( \cancel{400e^{-10t}} - \cancel{4000e^{-10t}} - 4000e^{-10t}t \right)$$

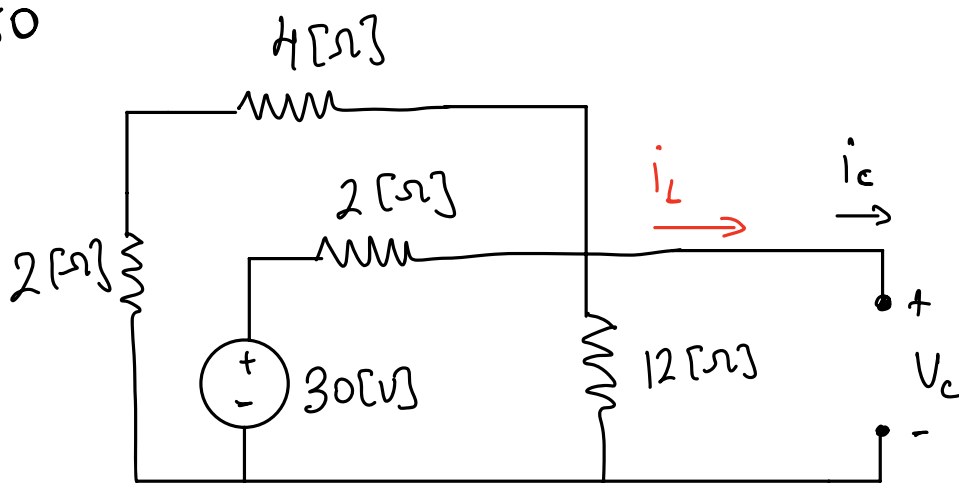
$$i_C(t) = -400t e^{-10t} \text{ [A]} \quad t \geq 0$$

\* **6.18** Determine  $i_C(t)$  in the circuit of **Fig. P6.18** for  $t \geq 0$ .



**Figure P6.18** Circuit for Problem 6.18.

$t < 0$

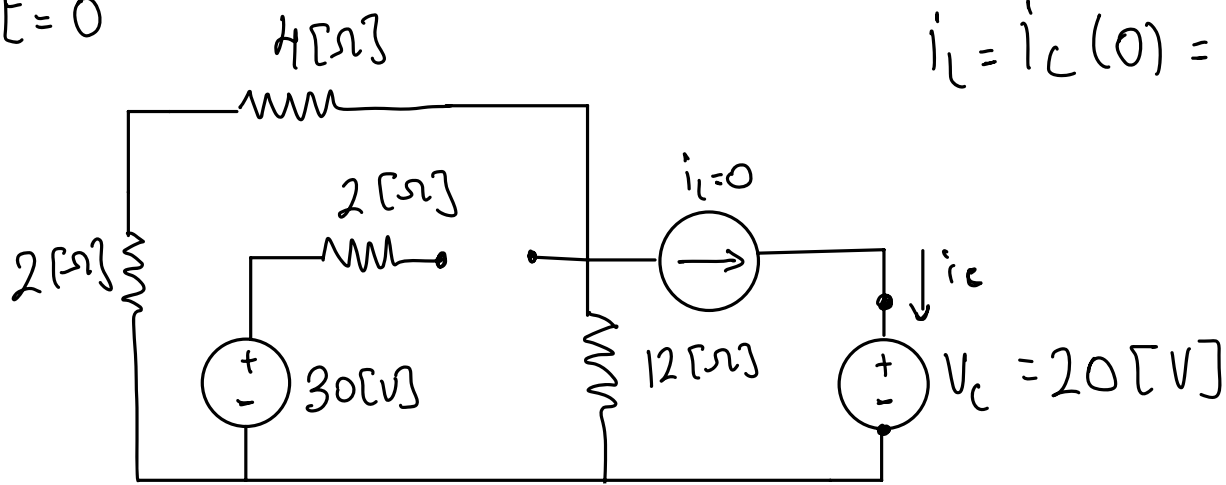


$$i_C(0^-) = 0[A] = i_L(0)$$

$$\frac{V_C}{12[\Omega]} + \frac{V_C - 30[V]}{2[\Omega]} + \frac{V_C}{6[\Omega]} = 0$$

$$V_C(0^-) = V_C(0^+) = 20[V]$$

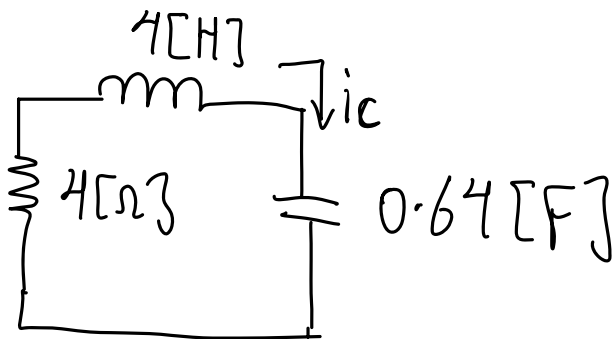
$$t = 0$$



2 and 4 are series  
and both of them are parallel  
with 12

$$t > 0$$

$$6 // 12 = 4[\Omega]$$



series RLC

$$\alpha = \frac{R}{2L}$$

$$= \frac{4}{2 \times 4} = 0.5[Np/s]$$

$$V_C(\infty) = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times 0.64}}$$

$$= 0.625 [rad/s]$$

$$\omega_0 > \alpha$$

so it is underdamped

$$V_C(t) = e^{-\alpha t} [D_1 \cos(\omega_d t) + D_2 \sin(\omega_d t)] + V_C(\infty)$$

$$D_1 = V_C(0) - V_C(\infty)$$

$$D_2 = \frac{\frac{1}{C} i_C(0) + \alpha [V_C(0) + V_C(\infty)]}{\omega_d}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{(0.625)^2 - (0.5)^2} = 0.375$$

$$D_1 = 20 [V]$$

$$D_2 = \frac{0.5(20)}{0.375} = 26.6$$

$$V_C(t) = e^{-0.5t} [20 \cos(0.375t) + 26.6 \sin(0.375t)]$$

$$i_C = C \frac{dV_C}{dt}$$

$$= -0.5 e^{-0.5t} [20 \cos(0.375t) + 26.6 \sin(0.375t)] + e^{-0.5t} [-7.5 \sin(0.375t) + 9.975 \cos(0.375t)]$$

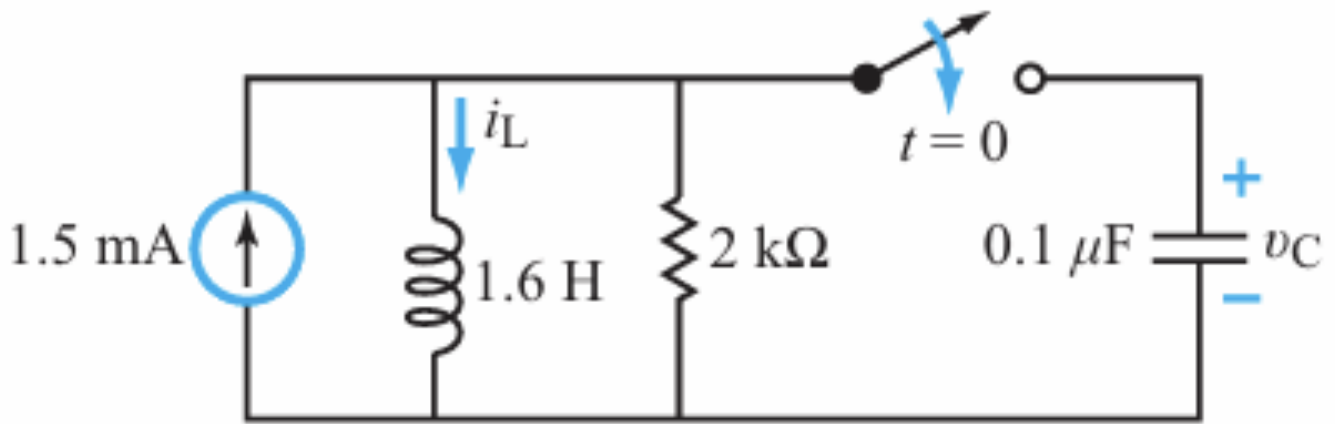
$$= -10 \cos(0.375t) e^{-0.5t} - 13.3 \sin(0.375t) e^{-0.5t} - 7.5 \sin(0.375t) e^{-0.5t} + 10 \cos(0.375t) e^{-0.5t}$$



$$i_c = 0.64 (-20.8 \sin(0.375t) e^{-0.5t})$$

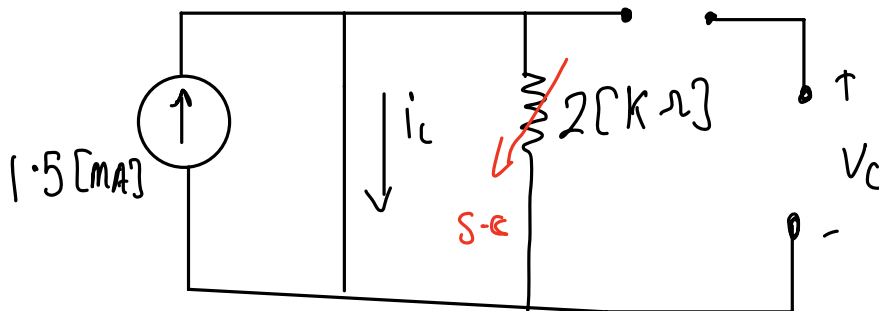
$$i_c(t) = -\frac{40}{3} \sin(0.375t) e^{-0.5t} \quad t \geq 0$$

**6.38** Determine  $i_L(t)$  in the circuit of **Fig. P6.38** and plot its waveform for  $t \geq 0$ . The capacitor had no charge on it prior to  $t = 0$ .



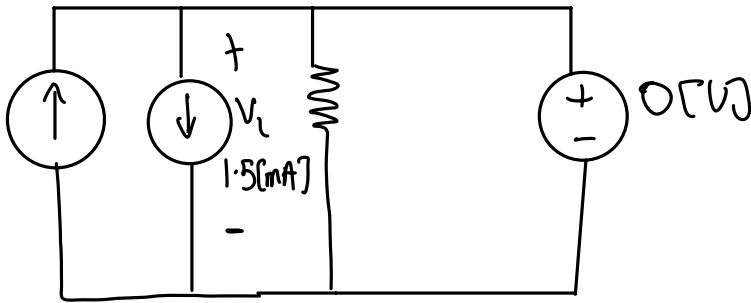
**Figure P6.38** Circuit for Problem 6.38.

$t < 0$



$$i_L(0^-) = i_L(0^+) = 1.5 \text{ [mA]} \quad V_C(0^-) = 0 \text{ [V]}$$

$$t = 0$$



$$V_L(0) = 0 \text{ [V]}$$

$$\alpha = \frac{1}{2RC} = 2500$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1.6 \times 10^{-1} \times 10^{-6}}} = 2500$$

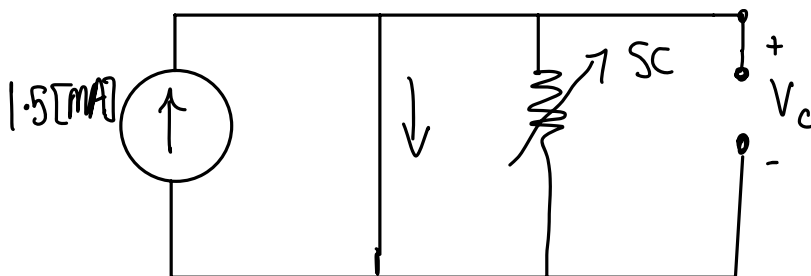
$$\alpha = \omega_0$$

critical damped

$$i_L(t) = (B_1 + B_2 t) e^{-\alpha t} + i_L(\infty)$$

$$B_1 = i_L(0) - i_L(\infty)$$

$$B_2 = \frac{1}{t} V_L(0) + \alpha [i_L(0) - i_L(\infty)]$$



$$i_L(\infty) = 1.5 \text{ [mA]}$$

$$\beta_1 = 0$$

$$\beta_2 = 0$$

$$i_L(t) = 1.5 \text{ [mA]}$$

$$t \geq 0$$